Chapter 4 Example

An ARMA(1, 2) model is fitted to a time series with 80 observations. The ACF coefficients of the residuals are shown below:

k	1	2	3	4	5	6	7	8
$r(k)$	0.14	0.08	−0.11	0.04	−0.13	0.05	−0.06	0.02

Based on the information above, perform diagnostic checking on the adequancy of the fitted model using the Ljung-Box statistic

$$LB = n(n+2)\sum_{k=1}^{m}\frac{r^2(k)}{n-k}$$

at 5% level of significance.

Solution:

H_0: Model is adequate $\qquad$ H_1: Model is not adequate

$$LB = 80(82)\left[\frac{0.14^2}{79}+\frac{0.08^2}{78}+\frac{(-0.11)^2}{77}+\frac{0.04^2}{76}+\frac{(-0.13)^2}{75}+\frac{0.05^2}{74}+\frac{(-0.06)^2}{73}+\frac{0.02^2}{72}\right]$$

$$= 6560\,[0.0002481 + 0.00008205 1 + 0.0000157 14 + 0.0000210526 + 0.0000225\dot{3} + 0.0000337844 + 0.0000049\dot{3}2 + 0.0000005]$$

$$= 6560\,(0.0008223315) = 5.3945$$

$df = 8 - 2 - 1 = 5$

Critical value: $\chi^2_{5;0.05} = 11.07$

Since $LB = 5.3945 < 11.07$, hence the test is not significant.

∴ The model is adequate